Physics 7510 Homework 5

Due Wednesday November 14

P1. In this problem we're going to explore the band structure of bilayer graphene. Usually bilayer graphene consists of two graphene sheets arranged such that the A sublattice in the bottom layer sits just below the B sublattice in the top layer; see Fig. 2. To describe the electronic properties of this system, let's introduce second quantized operators $c^\dagger_{A/B\mathbf{r}\sigma}$ that add electrons to the top layer and $d^\dagger_{A/B\mathbf{r}\sigma}$ that add electrons to the bottom layer. We can then write the Hamiltonian as a sum of three pieces,

$$H = H_{top} + H_{bottom} + H_{interlayer}. \tag{1}$$

The first two terms just describe nearest-neighbor hopping of electrons within each layer separately:

$$H_{top} = -t \sum_{\mathbf{r}\in A} \sum_{j=1}^{3} \sum_{\sigma} (c^\dagger_{A\mathbf{r}\sigma} c_{B\mathbf{r}+\hat{\mathbf{e}}_j\sigma} + h.c.) \tag{2}$$

$$H_{bottom} = -t \sum_{\mathbf{r}\in A} \sum_{j=1}^{3} \sum_{\sigma} (d^\dagger_{A\mathbf{r}\sigma} d_{B\mathbf{r}+\hat{\mathbf{e}}_j\sigma} + h.c.). \tag{3}$$

The third term allows electrons to tunnel between the two layers. Since the A sublattice in the bottom layer sits just below the B sublattice in the top layer, electrons tunnel most effectively between these sites. (The carbon p_z orbitals for these sites overlap most strongly.) Thus we can approximate $H_{interlayer}$ by

$$H_{interlayer} = -t' \sum_{\mathbf{r}\in A} \sum_{\sigma} (d^\dagger_{A\mathbf{r}\sigma} c_{B\mathbf{r}\sigma} + h.c.). \tag{4}$$

(a) How many sites are there per unit cell in bilayer graphene? Does the Brillouin zone change compared to that of single-layer graphene? If so, how?

(b) We'll find the band structure by going to momentum space. Express H in momentum space by using the Fourier transforms

$$c_{A/B\mathbf{r}\sigma} = \frac{1}{\sqrt{N_{uc}}} \sum_{\mathbf{k}\in BZ} e^{i\mathbf{k}\cdot\mathbf{r}} c_{A/B\mathbf{k}\sigma} \tag{5}$$

$$d_{A/B\mathbf{r}\sigma} = \frac{1}{\sqrt{N_{uc}}} \sum_{\mathbf{k}\in BZ} e^{i\mathbf{k}\cdot\mathbf{r}} d_{A/B\mathbf{k}\sigma}. \tag{6}$$

Although we essentially already did the transformation for H_{top} and H_{bottom}, repeat the steps to make sure you understand all the details. (It may be useful to use Mathematica to simplify some of the expressions.)

(c) It will be convenient to write the Hamiltonian as

$$H = \sum_{\mathbf{k}\in BZ} \sum_{\sigma} F^\dagger_\sigma(\mathbf{k}) \mathcal{H}(\mathbf{k}) F_\sigma(\mathbf{k}), \tag{7}$$

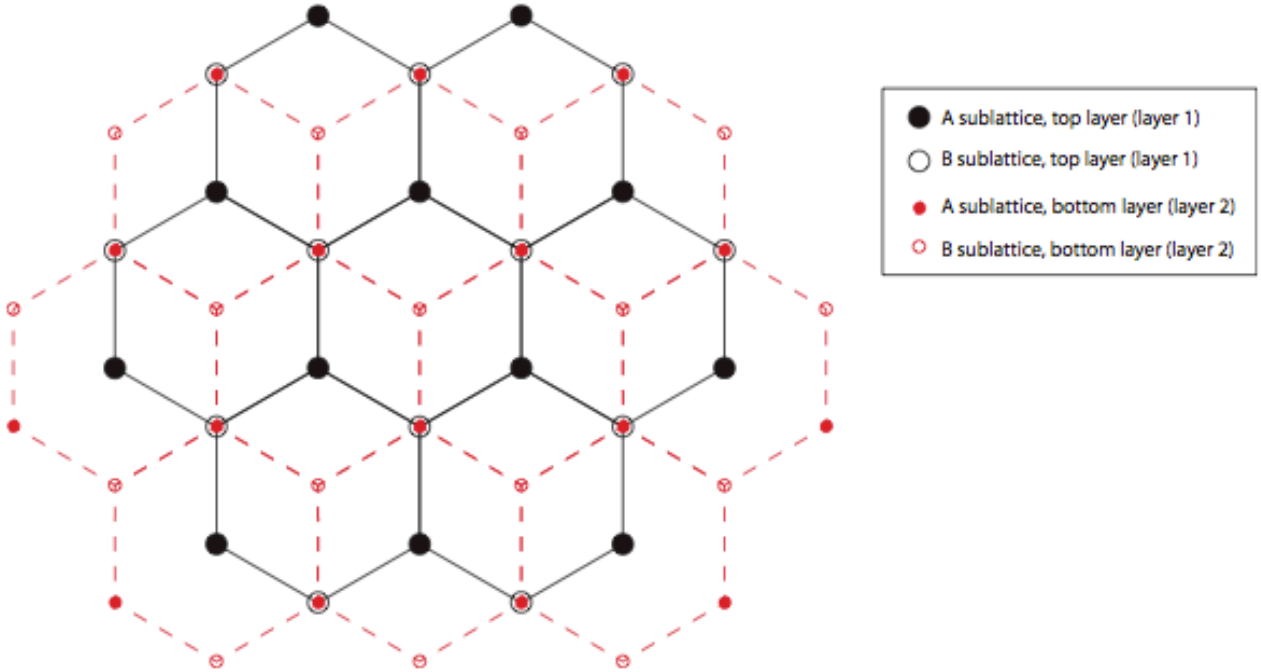

FIG. 1: Lattice structure of bilayer graphene

where

$$F_\sigma^\dagger(\mathbf{k}) = (c_{A\mathbf{k}\sigma}^\dagger \; c_{B\mathbf{k}\sigma}^\dagger \; d_{A\mathbf{k}\sigma}^\dagger \; d_{B\mathbf{k}\sigma}^\dagger). \tag{8}$$

Find the matrix $\mathcal{H}(\mathbf{k})$.

(d) The band energies are given by the eigenvalues of $\mathcal{H}(\mathbf{k})$. Find the band energies by diagonalizing $\mathcal{H}(\mathbf{k})$. Make a 3D plot of the band energies over the Brillouin zone when $t' = 0$ (using, say, Mathematica) to make sure that you see the Dirac cones in the spectrum. Then focus on $k_y = 0$ and plot the band energies as a function of k_x. In particular, as you 'turn on' t', illustrate how the band energies change near the Dirac points.

(e) A nice feature of bilayer graphene is that one can easily tune the electronic properties by applying a perpendicular electric field. This shifts particles from one layer to the other, which can be modeled by adding a chemical potential difference between the two layers via

$$\delta H = \delta\mu \sum_\sigma \left[\left(\sum_{\mathbf{r}\in A} c_{A\mathbf{r}\sigma}^\dagger c_{A\mathbf{r}\sigma} + \sum_{\mathbf{r}\in B} c_{B\mathbf{r}\sigma}^\dagger c_{B\mathbf{r}\sigma} \right) - \left(\sum_{\mathbf{r}\in A} d_{A\mathbf{r}\sigma}^\dagger d_{A\mathbf{r}\sigma} + \sum_{\mathbf{r}\in B} d_{B\mathbf{r}\sigma}^\dagger d_{B\mathbf{r}\sigma} \right) \right]. \tag{9}$$

By plotting the spectrum similar to what you did in part (d), deduce how bilayer graphene's band structure changes when $\delta\mu$ becomes non-zero.

P2. Pauli spin susceptibility of graphene. Here we focus on spin degrees of freedom and ask how Zeeman interaction $-\mu_B H$, where $\mu_B = e\hbar/(2mc)$ is the Bohr magneton, Dirac electrons of the graphene layer. It is convenient to start with thermodynamic potential

$$\Omega(\mu) = -k_B T \sum_{\lambda=\pm} \sum_{\sigma=\pm} \int \frac{d^2\vec{p}}{(2\pi\hbar)^2} \ln[1 + \exp\{\beta(\mu - \epsilon_{\lambda,\sigma}(p))\}],$$

where electron dispersion is $\epsilon_{\lambda,\sigma}(p) = \lambda v_F |\vec{p}| - \sigma\mu_B H$ and $\lambda = \pm$ is the band index (say, 'minus' for the lower band and 'plus' for the upper) while $\sigma = \pm$ is the spin projection on the direction of magnetic field H, and μ is the chemical potential. Once Ω is calculated with quadratic in H accuracy (see attached below pdf page with technical details), magnetization is found as $M = -\partial\Omega/\partial H$, and the spin susceptibility follows $\chi_{\text{Pauli}} = \partial M/\partial H$.
Use this approach to calculate Pauli susceptibility χ_{Pauli} for the neutral graphene ($\mu = 0$). Compare your result with that for the orbital (diamagnetic) susceptibility χ_{orb} calculated in the class, and estimate the ratio of the two at room temperature ($T \approx 300K$).

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(Follow Landau - Lifshitz, vol. 5, Ch. 59 "Magnetism of electron gas")

<u>Pauli susceptibility of graphene.</u>

Very generally, start with

expression without magnetic field

$$\Omega = -k_B T \left\{ \int \frac{d^2\vec{p}}{(2\pi\hbar)^2} \ln\left(1 + e^{\beta(\mu - E_-(p))}\right) + \ln\left(1 + e^{\beta(\mu - E_+(p))}\right) \right\}$$

this accounts for $E_\pm(p)$ bands; $\beta \equiv 1/k_B T$.

$E_\pm(p) = \pm v_F |\vec{p}|$ near K and K' points.

$$\Omega_0(\mu) = -g_v g_s k_B T \int \frac{d^2\vec{p}}{(2\pi\hbar)^2} \left\{ \ln\left(1 + e^{\beta\mu} e^{\beta v_F |p|}\right) + \ln\left(1 + e^{\beta\mu} e^{-\beta v_F |p|}\right) \right\}$$

valley degeneracy ($g_v = 2$)

spin degeneracy ($g_s = 2$)

In magnetic field electron's energy changes due to the Zeeman interaction $\pm \mu_B H$, $\mu_B = \frac{|e|\hbar}{2mc}$, and can be accounted for by shifting the chemical potential $\mu \to \mu \pm \mu_B H$ for electrons with different spins.

Then $\Omega(\mu) = \frac{1}{2}\Omega_0(\mu + \mu_B H) + \frac{1}{2}\Omega_0(\mu - \mu_B H)$

where Ω_0 is thermodynamic potential in the absence of the field.

For weak field, when $\mu_B H \ll k_B T$, we can Taylor expand: $\Omega_0(\mu \pm \mu_B H) = \Omega_0(\mu) \pm \frac{\partial \Omega_0}{\partial \mu}(\mu_B H) + \frac{1}{2}(\mu_B H)^2 \frac{\partial^2 \Omega_0}{\partial \mu^2}$

use in there

FIG. 2: Note on calculation of thermodynamic potential in the presence of the Zeeman magnetic